Unit 1 – Number Systems, Boolean Algebra & Logic Gates (Exam Notes)

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1. Number Systems (Types)

Number System	Base	Digits Used
Decimal	10	0–9
Binary	2	0, 1
Octal	8	0–7
Hexadecimal	16	0–9, A–F (A=10...F=15)

- **Binary**: computer ki language, sirf 0 aur 1. 4 bits = **Nibble**, 8 bits = **Byte**.
- **Decimal**: hum roz use karte hain. Position ka weight = power of 10.
  - Example: $2541 = (2\times10^3)+(5\times10^2)+(4\times10^1)+(1\times10^0)$
- **Octal**: 3 bits milke 1 octal digit banate hain.
- **Hex**: 4 bits milke 1 hex digit banate hain. A,B,C,D,E,F = 10,11,12,13,14,15.

Yaad rakhne wali trick: Binary→Octal = 3-3 bits group, Binary→Hex = 4-4 bits group.

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2. Number Base Conversions

(a) Binary → Decimal

Har bit ko uske positional weight (power of 2) se multiply karke add karo.

$(10110.001)2 = 1\times2^4+0\times2^3+1\times2^2+1\times2^1+0\times2^0 + 0\times2^{-1}+0\times2^{-2}+1\times2^{-3}$

$= 16+0+4+2+0+0+0+0.125 = (22.125)10$

(b) Binary → Octal

3-3 bits ke groups banao (dono taraf point ke), zeros add karke complete karo, fir octal digit likho.

111 110 101 011 . 001 100 → (7653.14)8

(c) Binary → Hexadecimal

4-4 bits ke groups banao, fir hex digit likho.

0111 1010 1011 . 0011 → (7AB.3)16

(d) Decimal → Binary/Octal/Hex

- **Integer part**: base se divide karte jao, remainders ko reverse order me likho.
- **Fractional part**: base se multiply karte jao, carry note karo (normal order me).

Example (152.25)10 → Binary:

152/2=76 rem 0 (LSB)

76/2=38 rem 0

38/2=19 rem 0

19/2=9 rem 1

9/2=4 rem 1

4/2=2 rem 0

2/2=1 rem 0

1/2=0 rem 1 (MSB)

→ Integer = 10011000

0.25×2=0.50 → 0

0.50×2=1.00 → 1

→ Fraction = .01

Answer: (152.25)10 = (10011000.01)2

Same method Octal (÷8) aur Hex (÷16) ke liye bhi.

(e) Octal → Decimal

Positional weight (power of 8) se multiply karke add karo.

(152.25)8 = $1\times8^2+5\times8^1+2\times8^0+2\times8^{-1}+5\times8^{-2}$ = 64+40+2+0.25+0.078 = 106.328

(f) Octal → Binary

Har octal digit ko 3-bit binary me likho.

(152.25)8 = (001 101 010.010 101)2

(g) Octal → Hex

Pehle Binary banao, fir 4-4 bits group karke Hex nikaalo.

(h) Hex → Decimal

Positional weight (power of 16).

(152A.25)16 = $1\times16^3+5\times16^2+2\times16^1+10\times16^0$+... = 5418.144

(i) Hex → Binary

Har hex digit ko 4-bit binary me likho.

(j) Hex → Octal

Pehle Binary banao, fir 3-3 bits group karke Octal nikaalo.

Shortcut yaad rakho: Binary hamesha "middle-man" hai — Octal↔Hex conversion binary se hoke hoti hai.

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3. Binary Arithmetic

Addition Rules

0+0=0

0+1=1

1+0=1

1+1=0 (carry 1)

Subtraction Rules

0-0=0

1-1=0

1-0=1

0-1=1 (borrow 1)

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4. Complements (Subtraction ka alternate tarika)

1's Complement

Sabhi 0 ko 1 aur 1 ko 0 kar do.

1010 → 0101

2's Complement

1's complement nikalo, fir usme **+1** add karo.

101 → 1's comp = 010 → +1 = 011 (2's complement)

Subtraction using 1's Complement (chhoti number ko badi se subtract karna)

Steps:

1. Chhoti number ka 1's complement nikaalo
2. Usse badi number me add karo
3. Jo final carry aaye usko result me **end-around carry** ke tarah add karo

1110 - 1001:

1110

+0110 (1's complement of 1001)

10100

carry 1 → add back

0100

\+ 1

0101 (Answer)

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5. Logic Gates

Gate = electronic circuit jo input signal (0/1) leke output signal deta hai.

2 Types:

- **Basic Gates**: AND, OR, NOT
- **Derived Gates**: NAND, NOR, XOR, XNOR

Gate	Symbol/Rule	Output HIGH kab?
AND (A.B)	dot	Sirf tab jab dono input 1 hon
OR (A+B)	plus	Jab koi bhi ek input 1 ho
NOT	bar	Input ko invert karta hai
NAND	AND + NOT	Jab koi bhi input 0 ho (sabse different: sab 1 ho to hi 0)
NOR	OR + NOT	Jab dono input 0 hon
XOR (A⊕B)		Jab dono inputs **different** hon (ek 0 ek 1)
XNOR		Jab dono inputs **same** hon

Truth Tables (yaad rakho, exam me common hai):

AND: 00→0, 01→0, 10→0, 11→1 OR: 00→0, 01→1, 10→1, 11→1 NAND: 00→1, 01→1, 10→1, 11→0 NOR: 00→1, 01→0, 10→0, 11→0 XOR: 00→0, 01→1, 10→1, 11→0 XNOR: 00→1, 01→0, 10→0, 11→1

Important line: NAND aur NOR ko **"Universal Gates"** kehte hain kyunki inse AND, OR, NOT sab bana sakte hain.

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6. Laws of Boolean Algebra

1. **Commutative Law**: order matter nahi karta
    - A.B = B.A , A+B = B+A
2. **Associative Law**: grouping matter nahi karta

- (A.B).C = A.(B.C) , (A+B)+C = A+(B+C)

3. **Distributive Law**:
   - A.(B+C) = A.B + A.C
   - A+(B.C) = (A+B).(A+C)
4. **AND Law**:
   - A.0=0 , A.1=A , A.A=A
5. **OR Law**:
   - A+0=A , A+1=1 , A+A=A

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7. De Morgan's Theorems (VERY IMPORTANT)

First Law:

(A.B)' = A' + B' (AND ka complement = OR of complements)

Second Law:

(A+B)' = A'.B' (OR ka complement = AND of complements)

Trick: "Bar todo, sign badlo" — jab bracket ke bahar ka bar todte ho, andar ka operator (. ya +) badal jaata hai.

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8. Adders

Half Adder

- 2 inputs (A, B), 2 outputs (Sum, Carry)
- Sum ke liye **XOR** gate, Carry ke liye **AND** gate

Sum = A XOR B

Carry = A AND B

Truth table:

A B | Sum Carry

0 0 | 0 0

0 1 | 1 0

1 0 | 1 0

1 1 | 0 1

Limitation: Carry-in nahi le sakta (sirf 2 numbers add kar sakta hai).

Full Adder

- 3 inputs (A, B, Cin), 2 outputs (Sum, Cout)
- Do half adders + 1 OR gate se banta hai

Step 1: A XOR B = partial sum

Step 2: (A XOR B) XOR Cin = final Sum

Cout = (A AND B) OR ((A XOR B) AND Cin)

Use: Multi-bit numbers (jaise 8-bit bytes) add karne ke liye, jaha carry ek stage se dusre stage me jaata hai.

Quick Revision (1-minute recap)

- 4 number systems: Decimal(10), Binary(2), Octal(8), Hex(16)
- Conversions: div-remainder (integer), multiply-carry (fraction); Binary is the bridge between Octal & Hex
- Binary Add/Sub rules + 1's & 2's complement (2's comp = 1's comp + 1)
- 7 Logic Gates: AND, OR, NOT, NAND, NOR, XOR, XNOR — NAND/NOR = Universal
- 5 Boolean Laws: Commutative, Associative, Distributive, AND law, OR law
- De Morgan's: (AB)'=A'+B' and (A+B)'=A'.B'
- Half Adder = XOR+AND (2 input); Full Adder = 2 Half Adders + OR (3 input, handles carry)